

Report - Version 4: Model Selection

Version 4 extends **Building Agentic AutoML** with automatic model selection.

The agent preserves task detection, feature detection, Data Inspection, and the two preprocessing strategies introduced in Version 3. The new capability is to compare **LightGBM, XGBoost, and CatBoost** under the same validation protocol and automatically select the best preprocessing-model combination.

Validation split, metrics, and hyperparameter tuning remain unchanged so that this version introduces only one new concept: **Model Selection**.

Goal of Version 4

- preserve the components introduced in Versions 1-3;
- build a small set of candidate models for classification and regression;
- evaluate every preprocessing-model combination on the same train-validation split;
- compare scores using the existing metric logic;
- store all experiments in the final state;
- return best_preprocessing, best_model, and best_score.

Architecture

```
Dataset + Target -> Task/Features -> Data Inspection -> Preprocessing Strategies -> Model  
Candidates -> Evaluation -> Best Experiment -> State
```

Task Detection

Task detection remains unchanged from Version 3: at most 20 unique target values means classification; otherwise the task is treated as regression.

```
def detect_task(df, target): n_unique = df[target].nunique() if n_unique <= 20: return  
"classification" return "regression"
```

Feature Detection

Feature detection is also unchanged. On Adult Income, the agent detects 6 numerical features and 8 categorical features.

Data Inspection

The Data Inspector introduced in Version 2 is preserved unchanged. On Adult Income it reports 48,842 rows, 15 columns, two target classes, dtypes, missing values, and cardinality. The main missing values remain workclass = 2,799, occupation = 2,809, and native_country = 857.

Preprocessing

Version 4 preserves the two strategies introduced in Version 3: native and impute. Imputation parameters are learned only from the training set. Some models require minimal technical adaptation of categorical features, without adding a new preprocessing concept.

Model Selection

This is the new capability introduced in Version 4. For each task, the agent builds three high-performance candidates: LightGBM, XGBoost, and CatBoost. No hyperparameter search is performed; default or near-default configurations are used to isolate the effect of model choice.

Metric Selection

Metric selection remains unchanged: ROC AUC for classification and RMSE for regression.

Training and Evaluation

Every preprocessing-model combination is evaluated using the same split, test_size=0.2 and random_state=42; classification also uses stratify=y. This keeps the comparison consistent with Version 3.

Results - Adult Income

Preprocessing	Model	ROC AUC
native	LightGBM	0.931166
native	XGBoost	0.929098
native	CatBoost	0.931383
impute	LightGBM	0.930688
impute	XGBoost	0.928546
impute	CatBoost	0.932105

Best experiment: **CatBoost + impute**, ROC AUC = **0.9321046183**.

Agent

- load dataset and target;
- detect task and feature types;
- inspect the dataset;
- prepare X and y;
- select the metric;
- build candidate models;
- try every preprocessing x model combination;
- select the best experiment;
- return the final state.

```
{ "task": "classification", "metric": "roc_auc", "best_preprocessing": "impute", "best_model":  
  "CatBoost", "best_score": 0.932104618262178, "experiments": [...] }
```

Regression Test

The same agent is executed on simple_regression.csv. It automatically detects regression, uses RMSE, and compares the same three models under both preprocessing strategies.

Preprocessing	Model	RMSE
native	LightGBM	1.876045
native	XGBoost	2.117240
native	CatBoost	1.789256
impute	LightGBM	1.877692
impute	XGBoost	2.113420
impute	CatBoost	1.654219

Best experiment: **CatBoost + impute**, RMSE = **1.6542186741**.

What Version 4 Teaches

Version 4 introduces the idea that an AutoML agent should not assume in advance which algorithm is best. It should compare compatible candidates under the same conditions and retain the evidence behind its choice.

The evolutionary pattern remains explicit: V1 builds the baseline, V2 observes the data, V3 experiments with preprocessing, and V4 compares models. The **One Version, One Concept** principle remains intact.

Limitations

- task detection still relies only on the number of unique target values;
- only two preprocessing strategies are available;
- only three boosting model candidates are compared;
- no cross-validation: evaluation still uses one train-validation split;
- no hyperparameter optimization;
- no feature engineering, planner, or multi-agent architecture;
- the state stores results and decisions, but not yet a persistent fitted pipeline.

Conclusion

Version 4 adds automatic model selection without changing the rest of the pipeline. The agent compares LightGBM, XGBoost, and CatBoost together with native/impute preprocessing, selects the best combination, and records all experiments. CatBoost + impute is the best configuration on both the classification example and the regression test. This version naturally prepares the project for a later evolution of the validation strategy.